

Subtractive Architecture: Minimizing Systemic Entropy via Structural Reduction

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Abstract

As logical frameworks expand, computational entropy and abstract architectural debt accumulate geometrically. Subtractive Architecture is the formalized mathematical doctrine of excising logic loops, dead compute dependencies, and probabilistic UI drifting over time. By aggressively applying thermodynamic constraints (Parsimony constraints) against monolithic repositories, we guarantee an immutable baseline of zero-variance hardware alignment, fundamentally achieving operational superiority by removing structural mass rather than adding complex heuristics.

1 Introduction

Sovereign macro-systems are plagued not by slow hardware, but by geometric abstraction layers. We postulate the "Logos Protocol Doctrine of Subtraction": any software architecture naturally converges towards chaos unless a destructive, downward mathematical pressure is applied. This thesis explores how structural reductionism algorithmically yields maximal application processing power.

2 Systemic Entropy Equations

Let the total system state dimensionality be bounded by Ω . Traditional architectural expansions attempt to resolve Ω by injecting abstract computational modules $\mathcal{M}$, inherently increasing the surface area of potential probabilistic logic failures. Excision fundamentally negates this vector.

Theorem 2.1 (Parsimony Minimization). *The highest localized throughput v_t of a computational cluster is achievable only when structural mass $M \rightarrow \infimum(S)$, defining S as the minimum viable logical axioms required for state progression.*

Proof. Let total structural stability $\mathcal{B}$ be inversely proportional to the volume of lines of code evaluated per execution subset. Removing abstraction directly destroys potential $O(N)$ execution divergence matrices. If the probability of function failure correlates strictly to its geometric footprint, subtraction structurally eradicates future error potentials symmetrically, rendering hardware-to-software execution latency essentially zero. $\square$

3 Sovereign Deployment Topology

By adopting Subtractive Architecture, modern UI clusters like Next.js and Firebase substrates shed monolithic caching mechanisms internally, routing all logic deterministically without middleware interference. This mathematical certainty is verified using strict AST (Abstract Syntax Tree) pruning rules and constant recursive audits against unused topological variables.

4 Conclusion

Subtractive Architecture confirms that within deterministic software boundaries, 'less' fundamentally does not equate to 'simple', but rather mathematically guarantees computationally perfect 'absolute'.

References
